

LMS Fee Logic – Final System Design Document

(Discounts, Installments, Auto-Debit, Fines, GST with Examples)

1) Core Fee Structure

Each student's total payable amount is calculated in this strict order:

1. Base Course Fee
 2. Course Starting Discount
 3. Personal/Admin Discount
 4. Progress Discount (after paying installments)
 5. Early Full Payment Discount (if applicable)
 6. GST applied at the end
 7. Installment plan generated
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2) Base Example

Course Fee = 40,000

Admission Fee = 5,000 (Non-refundable)

Remaining Fee = 35,000

GST = 18%

3) Discount Types

3.1 Course Starting Discount (Marketing Offer)

Example:

10% course discount

$40,000 - 4,000 = 36,000$

3.2 Personal Discount (Admin Approved)

Example:

Admin gives ₹2,000 for financial hardship

$36,000 - 2,000 = 34,000$

Store:

- reason
 - approvedBy
 - date
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3.3 Progress Discount (Reward)

Rule example:

- After paying 2 full installments → 5% discount on remaining amount

Example:

Remaining before discount = 25,000

5% of 25,000 = 1,250

New remaining:

$25,000 - 1,250 = 23,750$

Apply only once.

3.4 Early Full Payment Discount

If student pays full remaining amount early:

Example:

Extra discount = 1,000

4) GST Calculation

GST must be applied AFTER all discounts.

Example:

Final fee before GST = 33,000

GST 18% = 5,940

Total payable:

$33,000 + 5,940 = 38,940$

Important:

- GST applies to course fee portion

- Do NOT apply GST on:
 - Late fines
 - Penalties

5) Installment Plan Example

Total payable = 38,940

Installments = 6

Per installment:

6,490 each

6) Partial Payment Rule

Student must pay at least:

20% of installment

Example:

Installment = 4,000

Minimum to avoid fine = 800

Cases:

- Paid 0 → Fine applies
 - Paid 200 → Fine applies
 - Paid 800 → No fine
 - Paid 2,000 → No fine
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7) Late Fine System (Slab Based)

Installment = 4,000

Max fine allowed = 50% = 2,000

Days Late	Fine %	Fine Amount
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1–5 days	0%	0
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6–15 days	10%	400
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Days Late	Fine %	Fine Amount
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16–30 days	20%	800
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31–60 days	30%	1,200
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60+ days	50%	2,000 (cap)
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Fine never crosses 2,000.

8) Fine Conditions

Fine applies ONLY if:

- Paid amount < 20% of installment
AND
- Due date + grace days passed

If paid $\geq$ 20%:

No fine

9) Auto-Debit Logic

Auto debit tries:

- On due date
- After 5 days
- After 10 days

Admin decides retry count.

9.1 If full amount available

Installment marked PAID

9.2 If partial amount available

Example:

Installment = 4,000

Bank has 2,500

Auto debit = 2,500

Status = PARTIAL

If $\geq 20\%$ paid:

No fine

10) Payment Auto Allocation

Any payment goes in this order:

1. Old installment pending amount
2. Old fine amount
3. Current installment
4. Future installments

Example:

Student pays 6,000

Inst1 due = 4,000

Inst2 due = 4,000

System adjusts:

Inst1 = PAID (4,000)

Inst2 = PARTIAL (2,000)

No manual work.

11) Fine Still Payable After Installment Paid

Example:

Installment paid late.

Installment = 4,000

Fine = 1,200

Even if installment later becomes PAID:

Fine still pending = 1,200

Next payment must clear fine.

12) Course Access Rules

Before course end:

If dues pending:

- Warning shown

After grace:

- Block downloads

At course end:

If dues > 0:

- Block exam
 - Block certificate
 - Block course access
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13) Admission Fee Rules

- Mandatory payment
 - Non-refundable
 - Required to activate course access
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14) Safety Controls

Add system limits:

- Minimum partial percent = 20%
- Max late fine percent = 50%
- Max total discount allowed = 30% of course fee

Prevents misuse.

15) Student-Friendly Flexibility Options

Admin can:

- Give extra grace days
- Reduce fine %
- Give personal discount

- Pause auto-debit temporarily

Fields to store:

- fineWaiverPercent
- extraGraceDays
- paymentHoldTillDate

16) Automatic Installment Setup on Enrollment (Important Rule)

When a student enrolls in a batch, the system must automatically generate installment dates.

Rule:

If course duration = 6 months

Installments should be created only for 5 months.

Reason:

Last month is reserved for full payment clearance before course completion.

Example:

Course Start Date = Jan 1

Course Duration = 6 months

Course End = June 30

Installments auto-generated:

Installment Due Date

1	Feb 1
2	Mar 1
3	Apr 1
4	May 1
5	Jun 1

System does NOT create installment for last month.

17) 70% Course Completion Payment Rule

When course reaches 70% completion:

Student must clear full remaining balance.

Example:

Course duration = 6 months

70% = Around 4.2 months

So after 4th–5th month:

System checks:

if remainingAmount > 0

show alert: "Clear full fee before course completion"

If student does NOT clear:

After grace period:

- Apply penalty OR
- Remove any pending discounts
- Block exam eligibility
- Block certificate access

This forces final settlement before course ends.

18) Final Outcome

This system supports:

- Admission fee protection
- Multiple discount types
- GST handling
- Auto installment creation on enrollment
- 5-month installment plan for 6-month course
- 70% completion full-payment enforcement
- Partial payments
- Auto-debit retries
- Fine slabs
- Fine cap

- Reward discounts
- Payment auto-splitting
- Exam/certificate blocking

This is a complete production-level LMS fee engine.